Cooper C. Brown

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SUMMARY OF QUALIFICATIONS

I am aerospace engineering student designing solutions that require advanced integration of mechanical, electronic, and RF systems. I employ a variety of modeling and simulation software, especially in a reliably flight-systems context. I have experience utilizing project management and organization techniques to coordinate technical teams in achieving goals on time through rapid prototyping, and my passion for engineering of all disciplines ensures avoidance of issues during integration of subsystems.

EDUCATION

University of Maryland, A. James Clark School of Engineering — College Park, MD
Bachelor of Science in Aerospace Engineering — May 2027
Academic Honors & Aerospace Honors Program — Cumulative GPA: 3.85/4.00
Sigma Gamma Tau Aerospace Honors Society

SKILLS

Coding: Python, MATLAB, C++, C#, Rust, SQL, JavaScript, ROS2, Linux
CAD & Simulation: Ansys (Fluent, HFSS, Mechanical), SolidWorks, Fusion360, Simulink, KiCad, FluidX3D
Electronics: FPGAs, Arduino, ESP32, STM32, Raspberry Pi, through-hole/SMD Soldering, VNA
Mechanical Prototyping: FDM/SLA additive manufacturing, CNC routing, lathe work
Machine Learning (AI): Self-hosted Llama, Cuda & TensorRT, OpenCV
Certifications: Tripoli L2 High Power, Part 107 UAV Pilot, Student Private Pilot, Ham Radio General Class (KQ4TCN)

WORK EXPERIENCE

Army Research Laboratory — Adelphi, MD
National Security Scholars Intern — May 2025 – Present

- Rapidly prototyped sUAS aerostructures with DFM principles using SolidWorks and FEA for thin-wire landing
- Implemented OpenCV computer vision system for Python based optical flow wire-tracking and self-localization
- Field tested integrated systems for agile software development resulting 90% improvement in landing reliability

Georgia Tech Research Institute — Atlanta, GA
Engineering Research Intern — May 2024 – August 2024

- Developed and operated test hardware to validate novel quantum dot infrared sensor system
- Updated LabVIEW program to control spectrometer for variable wavelength sensor sensitivity characterization

PROJECT EXPERIENCE

University of Maryland X-Lab — College Park, MD
Autonomous Systems Researcher — May 2024 – Present

- Employed SolidWorks and Ansys CFD & FEA to optimize composite airframe for autonomous racing sUAS
- Integrated Nvidia Jetson with PX4 flight computer allowing for real-time edge inference of RL vision models

Terrapin Rocket Team — College Park, MD
Chief Engineer — June 2025 – Present

- Coordinated end-to-end development of sounding rocket subsystems for IREC 30,000 ft altitude competition
- Oversaw integration of aerostructures, propulsion, avionics, recovery, payload, and active aerobraking systems while targeting major team milestones and maintaining rapid iteration utilizing an agile development framework
- Simulated rocket dynamics using Ansys Fluent and MATLAB Simulink and conducted sub and full-scale test flights while gathering barometer, IMU, and GPS data to validate design decisions

Payload Team Lead — September 2024 – June 2025

- Lead design, fabrication, and testing of structural, electrical, and software systems for a robotic deployed parachute glider capable of autonomous navigation resulting in reliable landing at waypoint from a 1000 ft deployment altitude

Payload Engineer — August 2023 – September 2024

- Employed KiCad to design PCB and wrote flight software in C++ resulting in accurate positioning during high-altitude vehicle deployment from a sounding rocket